

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Computer Vision with OpenCV

COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO3: Analyze shape representation methods and techniques such as contours, regions, deformable models, and multi-resolution analysis. (BL4)

Assessment Tool Weightage: 30%

QUESTIONS: 2

Total Marks:30

Annexure A

Case study

Code of Conduct:

I B.Deepthi-192424269 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B.Deepthi

Case Study 1: Defect Detection in Industrial Parts

Case: A manufacturing company wants to automatically detect defects in gears and machine components using computer vision. The images are binary and may contain small protrusions or missing parts.

1(A) Region-Based Representation and Medial Axis

Region-based representation considers the complete area of an object rather than only its boundary. For a binary image, the component can be represented as:

Region-Based Representation

The binary image can be analyzed using connected-component analysis to identify each gear or machine component separately.

Important region features are:

- Area – total number of object pixels.
- Centroid – center position of the component.
- Bounding box – minimum rectangle containing the component.
- Perimeter – length of the boundary.
- Extent – ratio of object area to bounding-box area.
- Solidity – measures whether the region is solid or contains irregularities.

A defective component may have a different area or perimeter compared with a normal component.

For example:

A missing part generally decreases the area, while a protrusion may increase the area and perimeter.

Medial Axis

The medial axis represents the central structure of an object. It can be obtained by skeletonization or thinning.

The process is:

Binary Component → Skeletonization → Medial Axis → Shape Analysis

The medial axis helps identify:

- Structural changes
- Branches
- Missing portions
- Thickness variations
- Abnormal extensions

A missing tooth in a gear can cause a change or discontinuity in the expected skeleton. Similarly, an unwanted protrusion can create an abnormal branch.

Defect Detection

The detected shape can be compared with a normal reference component:

where represents the shape or skeleton.

If the difference exceeds a predefined threshold, the component can be classified as defective.

Conclusion

Region-based representation provides information about the size and geometry of the component, while the medial axis provides information about its internal structural shape. Combining both methods helps identify missing parts and unwanted protrusions accurately.

1(B) Effect of Noise and Preprocessing

Noise during image acquisition can seriously affect shape analysis. Since the system has to detect small protrusions and missing parts, even small unwanted pixels can be interpreted as defects.

Effects of Noise

Noise can produce:

1. False protrusions
2. Broken boundaries
3. Small unwanted regions
4. Holes inside components
5. Incorrect area measurement
6. Incorrect perimeter measurement
7. False contours
8. Unwanted branches in the medial axis

For example:

Preprocessing Steps

Step 1: Noise Filtering

Apply a suitable filter such as:

- Median filter
- Gaussian filter

A median filter is useful for removing isolated noise pixels.

Step 2: Thresholding

Convert the image into a clean binary image:

where T is the threshold.

Step 3: Morphological Opening

Opening removes small unwanted foreground objects:

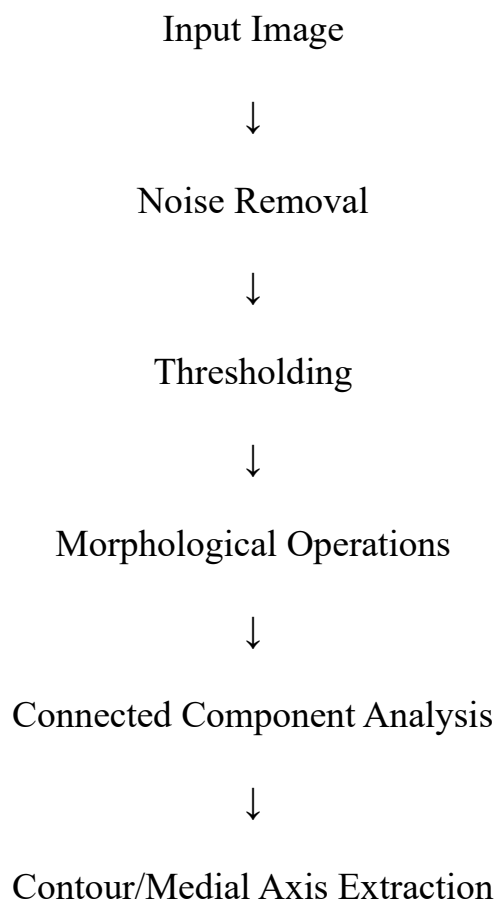
Step 4: Morphological Closing

Closing fills small gaps and holes:

Step 5: Connected Component Filtering

Very small components that are not part of the actual machine component can be removed according to their area.

Improved Processing Flow





Defect Detection

Conclusion:

Preprocessing is essential because noise can create false defects. Filtering, thresholding, morphological operations and connected-component filtering produce a cleaner image and improve the accuracy of subsequent shape analysis.

1(C) Fourier Descriptors vs. Wavelet Descriptors

Fourier descriptors and wavelet descriptors are techniques for representing the shape of objects. They can be used to classify industrial components as normal or defective.

Fourier Descriptors

Fourier descriptors represent the shape using frequency components of its boundary.

If a contour is represented as:

its Fourier transform can be calculated as:

The coefficients represent different frequency components of the shape.

Advantages

- Compact shape representation
- Good for overall/global shape
- Easy to compare similar components
- High-frequency coefficients can be reduced to suppress noise

Limitation

Fourier descriptors mainly describe the overall contour. Therefore, very small local defects may not be represented as effectively.

Wavelet Descriptors

Wavelet descriptors represent shape information at different resolutions or scales.

They can capture both:

- Global shape
- Local details

This is useful when a component contains a small protrusion or localized missing part.

Advantages

- Multi-resolution representation
- Good for local defects
- Captures sudden changes in shape
- Can provide information about where a defect occurs

Limitation

Wavelet-based analysis can be more complex than a simple Fourier representation.

Comparision

Global shape	Very effective	Effective
Local defects	Less effective	More effective

Small protrusions	Moderate	Very effective
Missing parts	Effective	Effective
Noise handling	Good	Good
Complexity	Relatively simple	Relatively higher
Best application	Overall shape classification Local + global defect detection	

Which is better for this case?

For large changes in the overall shape, Fourier descriptors are effective.

For small protrusions or localized missing parts, wavelet descriptors are generally more suitable because they preserve information at different scales.

Therefore, a combined feature representation can be used:

These features can then be given to a classifier to distinguish:

from

Conclusion

Fourier descriptors are strong for global shape representation, while wavelet descriptors are better for local and multi-scale shape changes. For the given industrial defect-detection problem, using both descriptors can provide a more robust solution.

Case Study 2: Medical Image Segmentation

Case: A hospital is developing a system to segment organs from CT scans. The organ boundaries are irregular and may vary slightly between patients.

2(A) Explain how active contours (snakes) can be applied to segment the organ boundaries.

Active contours, also called **snakes**, are deformable models used to detect and segment object boundaries in an image. They are suitable for CT images because organ boundaries may be irregular and can vary between patients.

1. Basic Representation

A contour can be represented parametrically as:

The contour is initially placed near the organ boundary and then gradually deformed toward the actual boundary.

2. Energy Function

The snake moves by minimizing an energy function:

Internal Energy

Internal energy controls the smoothness and flexibility of the contour.

where:

- controls elasticity.
- controls smoothness/curvature.

External Energy

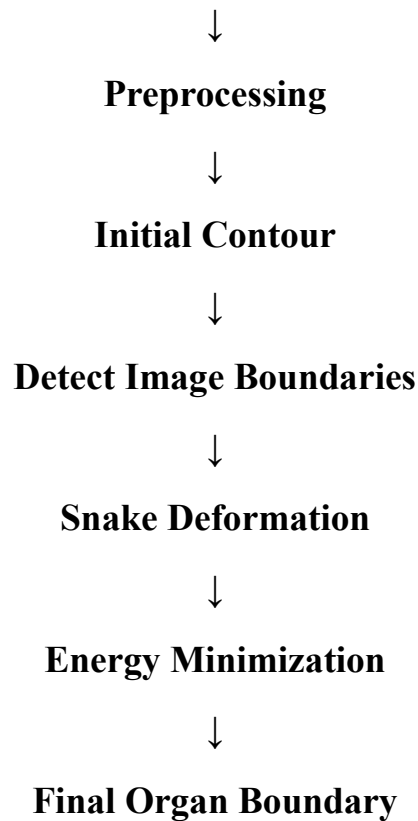
External energy attracts the contour toward strong image boundaries.

For example:

where represents the image gradient.

3. Segmentation Procedure

CT Image



4. Application

For example, if the approximate boundary of a liver is known, an initial contour is placed around it. The snake then moves toward the strong intensity changes corresponding to the liver boundary.

The contour continues to change until it reaches a stable position.

5. Advantages

- Can follow irregular boundaries.
- Produces smooth contours.
- Can adapt to variations in organ shape.
- Uses image information to locate boundaries.
- Reduces the need to manually trace the complete organ.

Conclusion

Active contours provide an effective deformable-model approach for organ segmentation. By balancing internal smoothness with external image forces, the contour can move toward and accurately represent the organ boundary.

2(B) Discuss the challenges of initializing contours and tuning parameters for accurate segmentation.

Initialization and parameter tuning are important because the performance of active contours depends on the **initial position of the contour and the selected parameters**. The case study specifically identifies these as challenges.

1. Contour Initialization

The initial contour should be reasonably close to the organ boundary.

If the contour is too far away:

- It may move toward an incorrect edge.
- It may fail to reach the organ.
- Segmentation accuracy decreases.

If the contour is too close or incorrectly positioned:

- Some portions of the organ may be missed.
- The contour may converge to an unwanted boundary.

Therefore:

2. Parameter Tuning

The important parameters include:

Elasticity

Controls how much the contour resists stretching.

If is too high:

and may not follow irregular boundaries.

If is too low:

and may become unstable.

Smoothness

Controls the curvature of the contour.

A high value produces a smoother contour but may remove important organ details.

A low value allows the contour to follow more detailed boundaries but can produce irregular results.

External Force

The external force attracts the contour toward image edges. If the organ boundary is weak, the snake may not be attracted strongly enough.

3. Challenges in CT Images

Medical CT images can contain:

- Noise
- Weak boundaries
- Similar intensity between neighboring tissues
- Irregular organ shapes
- Patient-to-patient variation

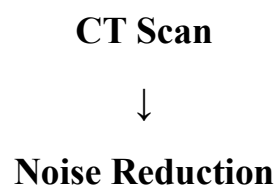
Therefore, the same initialization and parameters may not work for every patient.

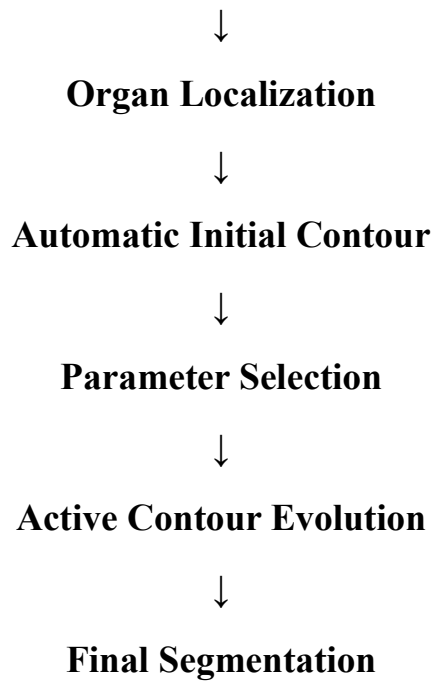
4. Proposed Solution

A better system can use:

1. Image preprocessing
2. Approximate organ localization
3. Automatic contour initialization
4. Appropriate parameter selection
5. Adaptive contour evolution
6. Post-processing

Flow





Conclusion

The major challenges are incorrect initialization and inappropriate parameter values. Automatic initialization and adaptive parameter tuning can improve segmentation accuracy and make the system more reliable for different patients.

2(C) How could level set methods improve segmentation for cases where organs split or merge in the images?

Level set methods are useful when organ regions **split or merge**, because they can handle changes in the topology of the contour. The assessment specifically asks how level sets can improve segmentation in such cases.

1. Basic Concept

Instead of representing the boundary directly as a parametric curve, a level set represents it using an implicit function:

The boundary is represented by the zero level set:

The function evolves until the desired organ boundary is obtained.

2. Handling Splitting

Suppose one segmented region needs to split into two regions:

A level set method can naturally represent this change during its evolution.

3. Handling Merging

Similarly, if two regions become connected:

the level-set representation can handle the topology change without explicitly creating or reconnecting contour points.

4. Level Set Evolution

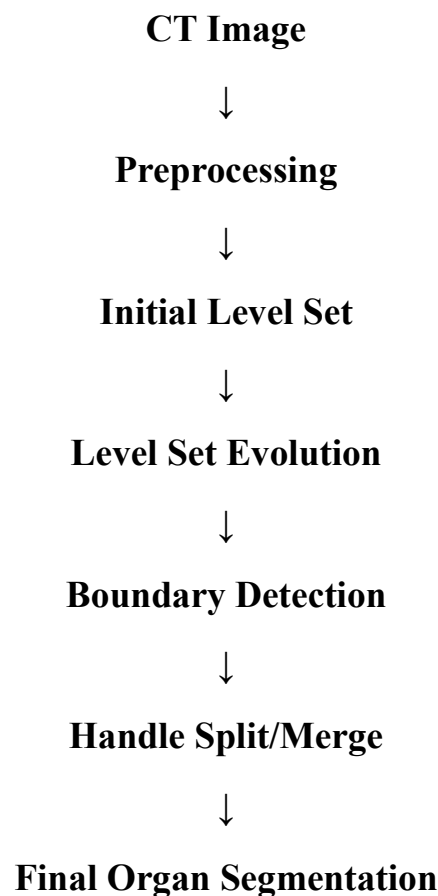
A simplified level-set equation is:

where κ represents the force controlling the evolution of the contour.

The level set evolves according to image information until the contour reaches the desired organ boundary.

5. Application to CT Segmentation

The processing can be represented as:



6. Advantages

- Handles irregular organ boundaries.
- Can naturally handle splitting.
- Can naturally handle merging.
- Supports complex topology changes.
- Does not require manually reconnecting contours.
- Suitable for complex medical-image segmentation.

7. Comparison with Active Contours

Feature	Active Contours	Level Set
Boundary representation	Explicit/parametric	Implicit
Irregular boundaries	Good	Good
Splitting	Difficult	Easy
Merging	Difficult	Easy
Topology changes	Limited	Strong
Complex organ shapes	Good	Very good

Conclusion

Level set methods provide a more flexible approach when organ boundaries **split, merge, or change topology**. Therefore, they can improve CT organ segmentation in complex cases where traditional active contours may have difficulty handling changing boundaries.